TP3 Assignment: MCS-based approach for end-to-end frame transmission reliability evaluation

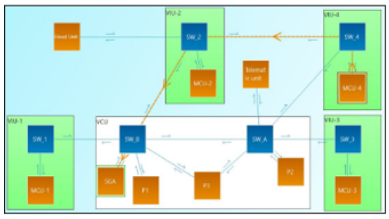
General description:

The following network architecture represents an automotive network, in which

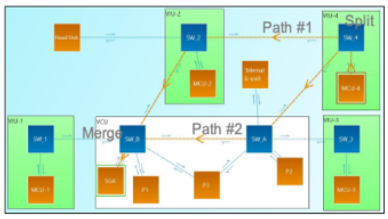
- Switches can be CB-aware or not
- Source and destination nodes can support static multi-homing or not.

Hence, packets can be transmitted from source to destination through the highlighted links. As we can see

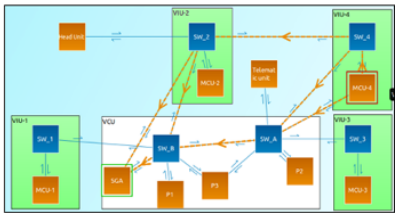
- There could be one or multiple paths from source to destination. Frame transmission on each link can fail or succeed.
- Packet transmission from source to destination will succeed when there is at least one valid path, i.e. all links involved in this path don't fail.



baseline topology



IEEE 802.1 CB



multi-home + IEEE 802.1 CB

For this assignment, you are required to use MCS (Minimal Cut Set)-based approach to conduct reliability analysis for different network designs. MCS-based reliability analysis can be performed in following steps:

1) Derive the Minimal Cut Sets corresponding to the top event, for example transmission between a selected pair of nodes fails (e. g. from source to destination), based on a given network topology;
2) Compute the failure probability of the top event (e.g. no transmission from source to destination), which can be approximated by summing up the failure probability of each individual MCS;
3) Compute the average time between packet loss based on the probability of the top-event;

Objective:

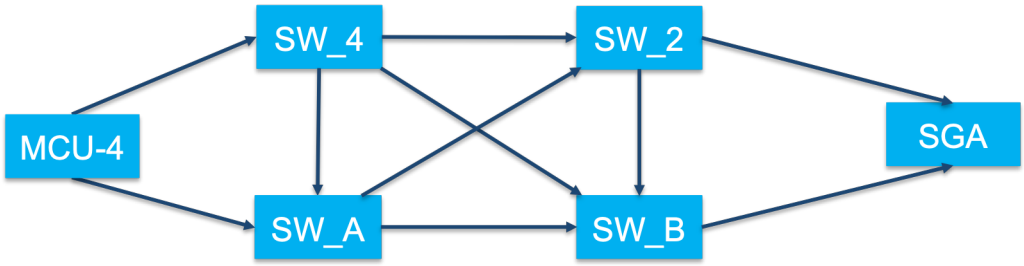
You need to develop a program in either Python or Java which is able to **derive MCSs for arbitrary network, for a selected pair of nodes**. And then compute reliability metrics such as failure probability of top event, average time between packet loss based on the derived MCSs.

The algorithm should then be applied to:

1) Derive MCSs and compute reliability metrics for the automotive network seen during the lecture, which supports different level of redundancy.

a. 3 different network architectures are considered: the baseline network architecture; the network with CB-aware switches only; the network with CB-aware switches and multi-homing end nodes.
b. 2 different bit error rates are considered: 10^{-10} and 10^{-12}.
c. It is assumed that the packet size is 3200 bits.
d. It is assumed that BER is the same for all links, bit errors are not correlated, and each bit has the same probability of being corrupted.

2) A network unseen in the lecture, which is shown below



Requirements:

- You need to develop using the template code provided, and implement at least the following 3 functions. You can add helper functions as you needed.
  1) **find_minimal_cut_sets(network, source, destination):** derive MCSs for the selected pair of nodes based on a given network topology.
  The function should return the list of derived MCSs, which will be used by the following functions.
  Hint: a minimal cut set is a combination of component failures that would lead to the top event. Removing any component failure from the set, the top-event will no longer occur.

  2) **derive_formula_for_top_event(list_of_mcs):** derive the formula to approximate the failure probability of the top-event based on the MCSs.

  3) **compute_reliability_metrics(list_of_mcs, ber):** compute the reliability metrics: failure probability of the top-event, and average time between packet loss, for the two BER rates

- As outputs, for each network (the 3 networks + the unseen network), your program must
  1) Output the Minimal Cut Sets derived for the top event, sorted in increasing order of cardinality of the MCSs.
  2) Output the formula which calculates the probability of the top event based on the MCSs. For example, following is the formula for the failure probability approximation of the top-event for network architecture with CB + dual-home.

```
6*q^2 + 2*q^3 + q^4
```

  Where q is a symbol representing the transmission failure probability of a packet over a link.

3) Output the probability of the top event and the average time between two packet losses (in hour), for the two Bit Error Rate (BERs). It is assumed that packet is sent every 10ms.

Notes:

- You can use either Python or Java for the implementation. If Python is used, the latest version is preferred. For Java, you can derive your own template from the python template.
- Python decimal package is needed for higher precision. You need to find the equivalent package for Java.
- Implement your own MCS algorithm, use of any existing library for deriving MCS is strictly forbidden. If doing so, the submission will automatically get 0 point.
- **Work in group of 2-3 people**, members in the same group will get the same score. One submission to Moodle is sufficient per group, this can be done by either group member. **Don't make duplicate submissions** to Moodle.
- To give you more time to work on the project, there will be **no class on Nov. 4th**.

Submissions: submit your **python/java code**, **the slides in PDF and the link of the presentation video to Moodle by Nov. 13th, 23:59**. This is a firm deadline, no extension will be made. Late submission will not be accepted and will be automatically rejected.

Code:

- Add suitable comments to your code to ease understanding. Code without clear comments will automatically receive a 15% of the overall score as penalty.

Slides:

- **Important: indicate all the group members in the title page of your slides.**
- 10 – 20 slides, including the title page. No need to prepare lengthy slides. Presentation and clarity are more important.
- The slides should be well-formatted and self-contained, which means that it can be fully understood without opening and running your program.
- Presentation of the slides is also part of the evaluation.
- In the slides, you need to at least include
  1) A clear explanation to how the 3 required functions are realized, in particular how to find the MCSs. **Explain the idea behind**, rather than the code line-by-line. To better support explanation, you can include figures, diagrams, formula, illustrating example, code snippet, etc.
  2) One slide for each network (baseline, CB-aware, CB aware + dual-home, unseen), which presents the analysis result for each network, including:
     a. A figure corresponding to the network
     b. The Minimal Cut Sets found by the program
     c. The formula derived for the probability of the top-event by the program
     d. The reliability metrics (failure probability of the top event, average time between packet loss) computed for each BER

3) A comparison table for the 4 networks on a separate slide, including the MCSs found for each network, the failure probability of the top event, the average time between packet loss. You can use the results for BER=10^{-10}.
4) One slide that performs qualitative analysis **based purely on the MCSs** for each network. For each network, you need to answer:
    a. Whether the network has any single-point of failure;
    b. What is the maximum number of simultaneous link failures that the network can tolerate: any single-link failure, any combination of two link failures, etc.

    Justify your answers based on the MCSs.
5) **One slide which explicitly recognizes the contribution of each group member**. Each group member should be involved in at least one activity, otherwise he/she will receive 0 point for this assignment.

Presentation video:

- A 10-15 minutes video presentation based on the prepared slides. Presentation <8min or >20min will receive a 10% of the overall score as penalty.
- The presentation needs to contain **2 parts**: first the **presentation of the slides**, second a **demo of the running program**.
- Use your preferred tool for video and screen recording. The presenter(s) must keep their camera(s) on and remain visible in the recording. The presentation can be delivered by one or multiple group members.
- Include a clickable link to your presentation video on the first slide, in addition to submitting it on Moodle. The link should remain accessible. If the video is downloadable, it is even better.
- Clarity of the presentation is also part of the evaluation.

Evaluation:

- The assignment is graded out of 20 points, and it is a weighted sum of the 3 parts. The weights for the different parts are: **50%** for the code, **25%** for the slides; **25%** for the presentation.
- If either the slides or the presentation video is missing in the submission, the code will not be considered, the submission will automatically get 0.